

richcast: Hindcast Species Distributions and Assemblage Richness Through Time

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Title Hindcast Species Distributions and Assemblage Richness Through Time

Version 0.1.0

Description Fits a species distribution model for every species near a region, projects it onto palaeoclimate reconstructions across a series of time slices, and stacks the results into assemblage richness surfaces. Ranges come from IUCN Red List polygons supplied by the user; each model is an equally weighted ensemble of a random forest and MaxEnt, trained on pseudo-presences sampled from the range polygon and evaluated by AUC and the continuous Boyce index. Climate data are retrieved via 'pastclim'. A trait table for rodents from Ecke et al. (2022) [doi:10.1038/s41467-022-35273-7](https://doi.org/10.1038/s41467-022-35273-7) is bundled as a worked example.

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Contents

bioclim_vars	2
bp_to_ce	3
build_taxon_db	4
check_climate_grids	5
check_climate_products	6
climate_dir	6
filter_taxa	8
fit_sdm	9
iucn_folder	11
normalise_species	12
pastclim_climate	13
prepare_climate	14
presence_thresholds	15
project_sdm	16
region	17
region_presets	17
richness_at	18
richness_grid	19
richness_in	20
richness_surface	21
rodent_traits	21
run_hindcast_series	22
sf_polygons	24
species_near	25
suitability	26
suitability_grid	26
zooarch_taxa	27

bioclim_vars

The eight canonical bioclimatic predictors

Description

Every model is fitted on these, or a subset of them: annual mean temperature (bio01), temperature seasonality (bio04), maximum temperature of the warmest month (bio05), minimum temperature of the coldest month (bio06), annual precipitation (bio12), precipitation seasonality (bio15), and precipitation of the wettest (bio16) and driest (bio17) quarters. Together they describe the means and extremes of temperature and moisture while leaving out the more redundant bioclim variables, and all eight are available in the common palaeoclimate reconstructions (Beyer2020 has no bio02 or bio03, for example).

Usage

```
bioclim_vars
```

Format

A character vector of length 8.

Examples

```
bioclim_vars
```

bp_to_ce

Convert between years BP and years CE

Description

richcast works in years CE throughout, because a single signed axis orders correctly whether a study spans the Holocene or the last millennium. Deep time is conventionally quoted in years before present, though, and "present" in that convention means 1950 CE – so palaeoclimate reconstructions land on years like 1950, 950, -50, -1050 rather than round thousands.

Usage

```
bp_to_ce(bp, present = 1950)
```

```
ce_to_bp(ce, present = 1950)
```

Arguments

bp, ce	Numeric vectors of years before present / years CE.
present	Year CE that BP counts back from. 1950 by convention.

Details

These convert, so the offset is written down once instead of being rediscovered as an off-by-1950 error.

Value

A numeric vector.

Examples

```
# The Last Glacial Maximum, roughly 20 ka BP:
bp_to_ce(-20000)

# Beyer2020's millennial slices, as richcast wants them:
bp_to_ce(seq(0, -10000, by = -1000))

ce_to_bp(1950)
```

build_taxon_db	<i>Assemble a taxon database from ranges and (optionally) traits</i>
----------------	--

Description

Joins a resolved range source to a trait table and returns the object every other richcast function consumes.

Usage

```
build_taxon_db(
  ranges,
  traits = NULL,
  trait_species_col = "species",
  dissolve = TRUE,
  crs = 4326,
  quiet = FALSE
)
```

Arguments

ranges	A range source from iucn_folder() , iucn_shapefile() or sf_polygons() ; a bare sf object; or a folder path, which is read with iucn_folder() defaults.
traits	Optional data frame of species attributes, e.g. rodent_traits . Joined on normalised species name.
trait_species_col	Name of the species column in traits.
dissolve	Logical. Union multiple features of the same species into a single geometry. Defaults to TRUE.

crs	Target coordinate reference system. Defaults to EPSG:4326.
quiet	Suppress progress messages.

Details

Two behaviours are worth knowing about, because both silently corrupted results in the pipeline this package was extracted from:

- **Multi-row species are dissolved.** Range databases routinely store one species across many features – IUCN splits by subspecies, seasonality and disjunct patches, so a global rodent download holds ~3100 features for ~2350 species. Treating each feature as a species means a model is trained against a single fragment while the rest of the species' true range is handed to the background sample, i.e. the model learns to discriminate the species from itself. Set `dissolve = FALSE` only if you genuinely want per-feature modelling.
- **The trait join is reported, not silent.** Species present in the ranges but absent from traits keep their geometry and receive NA traits. The count is printed, because an inner-join-shaped drop is easy to mistake for a real biological filter.

Value

A `richcast_db`: an `sf` object with a species column first, trait columns next, and geometry last.

See Also

[iucn_folder\(\)](#), [rodent_traits](#)

Examples

```
ranges <- sf::st_sf(
  species = c("Genus_alpha", "Genus_alpha", "Genus_beta"),
  geometry = sf::st_sfc(
    sf::st_polygon(list(cbind(c(0, 1, 1, 0, 0), c(0, 0, 1, 1, 0)))),
    sf::st_polygon(list(cbind(c(2, 3, 3, 2, 2), c(0, 0, 1, 1, 0)))),
    sf::st_polygon(list(cbind(c(1, 3, 3, 1, 1), c(1, 1, 3, 3, 1)))),
    crs = 4326
  )
)
# The two Genus_alpha features collapse into one range.
build_taxon_db(sf_polygons(ranges), quiet = TRUE)
```

check_climate_grids	<i>Verify that every prepared slice shares one grid</i>
---------------------	---

Description

A silent resample between the fitting grid and the projection grid is one of the easier ways to get confident nonsense out of an SDM, so this compares geometry with zero tolerance and fails loudly.

Usage

```
check_climate_grids(climate, vars = NULL, quiet = FALSE)
```

Arguments

climate	A <code>climate_dir()</code> source.
vars	Variables that must be present in every slice.
quiet	Suppress the success message.

Value

TRUE, invisibly, or an error.

```
check_climate_products
```

Warn when the present slice and the time slices come from different products

Description

Fitting on one climate product and projecting onto another folds the step between them into every `delta_from_present`. Measured on one species, that step was 42% of its present-day range and flipped it from above-present in 2 of 11 centuries to above-present in all 11 – and the offset is species-specific in sign, so it does not cancel.

Usage

```
check_climate_products(climate, quiet = FALSE)
```

Arguments

climate	A <code>climate_dir()</code> source.
quiet	Suppress the all-clear message.

Details

Only works where the slices were prepared by `prepare_climate()`, which leaves a manifest. Directories assembled by hand are unreadable in this respect and pass silently.

Value

TRUE if consistent (or undeterminable), FALSE if mixed.

climate_dir

Climate source: a directory of pre-downloaded rasters

Description

Expects one subdirectory per time slice, each holding one GeoTIFF per variable named after that variable (bio01.tif, bio12.tif, ...). This is the layout written by [prepare_climate\(\)](#).

Usage

```
climate_dir(path, present = "present_1985", slice_fmt = "time_%04d")
```

Arguments

path	Directory holding the time-slice subdirectories.
present	Subdirectory name for the slice models are fitted against.
slice_fmt	sprintf() template mapping a year CE to a subdirectory name. The default zero-pads to four digits, matching prepare_climate() .

Details

```
climate/
  present_1985/  bio01.tif bio04.tif ...
  time_0850/    bio01.tif bio04.tif ...
  time_1850/    bio01.tif bio04.tif ...
```

All slices must share a grid. This is checked on first use rather than assumed, because a mismatch between the fitting grid and the projection grid produces plausible-looking but meaningless suitability surfaces.

Value

A `richcast_climate`.

Choosing the fitting slice

`present` names the directory a model is fitted against, and it need not be an observational product. Fitting on WorldClim and projecting onto CHELSA means the model crosses datasets between fitting and projection, so any systematic difference between the two products is indistinguishable from climate change. Where the palaeoclimate reconstruction publishes its own present-day slice, fitting on that instead keeps one product throughout:

```
climate_dir("climate", present = "chelsa_1950")
```

The trade-off is that a reconstruction's present-day slice is itself modelled, so it inherits the model's biases rather than an observational network's. Neither choice is free; running both and comparing is cheap once the slices are on disk.

BCE slices

Years are signed. `sprintf("time_%04d", -1050)` gives `time_-1050`, keeping 1050 BCE in its own directory rather than colliding with 1050 CE.

Examples

```
clim <- climate_dir("Inputs/Climate/eurasia_slices")
clim
```

filter_taxa	<i>Filter a taxon database</i>
-------------	--------------------------------

Description

A thin, self-documenting wrapper around `dplyr::filter()` that reports how many species each expression removed. Useful because trait filters often remove far more (or far less) than intended – for example `S_index > 0` on `rodent_traits` looks like a risk threshold but every matched species scores above zero, so it only ever removes species missing from the trait table.

Usage

```
filter_taxa(db, ..., quiet = FALSE)
```

Arguments

<code>db</code>	A <code>richcast_db</code> .
<code>...</code>	Filter expressions, passed to <code>dplyr::filter()</code> .
<code>quiet</code>	Suppress the report.

Value

A filtered `richcast_db`.

Examples

```
db <- build_taxon_db(
  sf_polygons(sf::st_sf(
    species = c("Genus_alpha", "Genus_beta"),
    mass = c(10, 500),
    geometry = sf::st_sfc(
      sf::st_polygon(list(cbind(c(0, 1, 1, 0, 0), c(0, 0, 1, 1, 0)))),
      sf::st_polygon(list(cbind(c(2, 3, 3, 2, 2), c(0, 0, 1, 1, 0)))),
      crs = 4326
    )
  )),
  quiet = TRUE
)
filter_taxa(db, mass < 100)
```

fit_sdm

*Fit a species distribution model***Description**

Fits an ensemble of a random forest (via **ranger**) and MaxEnt (via **maxnet**) for one species against present-day climate, and returns an object that `project_sdm()` can push onto any number of palaeo-climate slices.

Usage

```
fit_sdm(
  db,
  species,
  climate,
  predictors = bioclim_vars,
  n_presence = 100,
  n_background = 1000,
  buffer = 0.3,
  threshold = "p10",
  test_frac = 0.25,
  num_trees = 500,
  seed = 123,
  land = NULL,
  quiet = FALSE
)
```

Arguments

db	A richcast_db from <code>build_taxon_db()</code> .
species	Species name, or a row index into db.
climate	A climate source from <code>climate_dir()</code> or <code>pastclim_climate()</code> .
predictors	Climate variables to fit on: the eight <code>bioclim_vars</code> by default, or a subset of them.
n_presence	Number of pseudo-presences drawn inside the range.
n_background	Number of background points drawn outside it.
buffer	Study-extent buffer around the range's bounding box, as a proportion of the box's diagonal.
threshold	"p10", or a fixed number in (0, 1).
test_frac	Share of presences and background held out for evaluation.
num_trees	Trees in the random forest.
seed	Random seed for sampling, the hold-out split and the forest.
land	Optional sf/sfc land outline used to mask predictions and restrict the background to land. Defaults to Natural Earth at medium resolution.
quiet	Suppress progress messages.

Value

A `richcast_sdm.metrics` holds AUC and Boyce for maxent, rf and ensemble; `range_cells` is the number of grid cells with climate the range covers.

Training data

The study extent is the range polygon's bounding box, widened on every side by `buffer` times its diagonal (30% by default). Inside it, `n_presence` pseudo-presences are drawn from grid cells inside the range polygon and `n_background` background points from land cells outside it, both without replacement. A range covering fewer cells than `n_presence` contributes every cell it has; `range_cells` on the result says how many that was.

Members and ensemble

- **MaxEnt**: `maxnet::maxnet()` with default features and `regmult = 1`, predicted on the cloglog scale.
- **Random forest**: a probability forest from `ranger::ranger()`, with each tree drawn from an equal number of presences and background points (balanced down-sampling), which keeps the 1:10 class imbalance from flattening predicted probabilities toward zero.
- **Ensemble**: the unweighted mean of the two, so it is also on `[0, 1]`.

The presence/absence threshold is applied to the ensemble only. The default "p10" is the tenth percentile of ensemble predictions at the training presences: it tolerates 10% omission, and being referenced to the presences it does not tighten as the background widens.

Evaluation

Before the final fit, a stratified `test_frac` of the presences and background is held out, both members are trained on the rest, and each member and the ensemble are scored on the held-out points:

- **AUC**, the probability that a held-out presence scores above a held-out background point.
- **Continuous Boyce index** (Hirzel et al. 2006): the Spearman correlation between suitability and the predicted-to-expected ratio of held-out presences, over a moving window. Near 1 means presences concentrate where the model says suitability is high; near 0 means no better than random.

The final members are then refitted on all the points. Set `test_frac = 0` to skip evaluation.

References

Hirzel, A. H., Le Lay, G., Helfer, V., Randin, C., & Guisan, A. (2006). Evaluating the ability of habitat suitability models to predict species presences. *Ecological Modelling*, 199, 142-152.

See Also

[project_sdm\(\)](#), [run_hindcast_series\(\)](#)

iucn_folder

Range source: IUCN Red List range polygons

Description

iucn_folder() reads every shapefile (and geopackage) found anywhere under a folder, so a set of Red List downloads unpacked side by side – say bovid_IUCN/, cervid_IUCN/ and equid_IUCN/ – is read as one source. iucn_shapefile() reads a single file.

Usage

```
iucn_folder(
  path,
  species_col = "SCI_NAME",
  species = NULL,
  presence = 1:3,
  origin = 1:2
)
```

```
iucn_shapefile(
  path,
  species_col = "SCI_NAME",
  species = NULL,
  presence = 1:3,
  origin = 1:2,
  layer = NULL
)
```

Arguments

path	For iucn_folder(), a folder searched recursively. For iucn_shapefile(), a .shp or .gpkg file.
species_col	Name of the binomial column. IUCN exports call this SCI_NAME.
species	Optional character vector of binomials. When given, only these are read, via an OGR attribute query; names match in both Genus species and Genus_species form.
presence, origin	IUCN PRESENCE and ORIGIN codes to keep, or NULL for all. See details.
layer	Optional layer name, for multi-layer sources.

Details

The data are **not** redistributed by this package and are not bundled with it: the IUCN Red List Terms of Use (v3, section 4) prohibit redistribution of Red List data, whole or in part, including within derivative works. Download the ranges yourself from <https://www.iucnredlist.org/resources/spatial-data-download>, accept the terms, and cite the version you used.

Value

A richcast_range_source.

Which polygons count as today's range

Red List files carry historical and uncertain range alongside the current one. By default only polygons coded as extant (PRESENCE 1-3: extant, probably extant, possibly extant) and native or reintroduced (ORIGIN 1-2) are kept, since an extinct or introduced patch is not part of the climate niche a model should learn. Set either argument to NULL to keep every code. Files without the column are not filtered on it.

See Also

`build_taxon_db()`, `sf_polygons()`

Examples

```
src <- iucn_folder("UngulatePolygons")
print(src)

# Only the taxa you actually intend to model:
iucn_shapefile(
  "~/iucn_artiodactyla/data_0.shp",
  species = c("Gazella gazella", "Sus scrofa", "Capra ibex")
)
```

normalise_species	<i>Normalise species names to a canonical form</i>
-------------------	--

Description

Range databases and trait tables rarely agree on binomial formatting, and a silent mismatch here shows up much later as an unexplained drop in species count. This applies one canonical form to both sides of the join: whitespace-trimmed, internal whitespace collapsed to a single underscore, and sentence case (Genus_species).

Usage

```
normalise_species(x)
```

Arguments

x Character vector of species names.

Value

A character vector of the same length.

Examples

```
normalise_species(c("Marmota baibacina", "marmota_bobak", " Rattus rattus "))
```

pastclim_climate	<i>Climate source: live pastclim queries</i>
------------------	--

Description

Retrieves slices through **pastclim** on demand. Convenient for exploration; for a full run over many species and time slices, prepare the slices once with [prepare_climate\(\)](#) and use [climate_dir\(\)](#) instead, since every species would otherwise re-stream the same rasters.

Usage

```
pastclim_climate(
  dataset_present = "WorldClim_2.1_5m",
  dataset_past = "CHELSA_trace21k_1.0_0.5m_vsi",
  present_time = 1985,
  times = NULL
)
```

Arguments

dataset_present, dataset_past	pastclim dataset names.
present_time	Year CE treated as "present" when fitting.
times	Numeric vector of years CE this source can serve.

Value

A `richcast_climate`.

Examples

```
clim <- pastclim_climate(times = seq(850, 1850, by = 100))
clim
```

prepare_climate

*Download and prepare climate slices for hindcasting***Description**

Streams climate rasters through **pastclim**, aggregates them to a common grid, and writes one Geo-TIFF per variable per time slice in the layout `climate_dir()` expects.

Usage

```
prepare_climate(
  path,
  vars = bioclim_vars,
  times,
  extent,
  present_time = 1950,
  dataset_past = "CHELSA_trace21k_1.0_0.5m_vsi",
  dataset_present = dataset_past,
  agg_past = 20,
  agg_present = agg_past,
  present_dir = "present",
  slice_fmt = "time_%04d",
  check_grids = TRUE,
  quiet = FALSE
)
```

Arguments

path	Output directory. One subdirectory per slice is created.
vars	Bioclimatic variables to prepare: the eight bioclim_vars by default, or a subset of them.
times	Numeric vector of years CE for the palaeoclimate slices.
extent	Numeric c(xmin, xmax, ymin, ymax) to clip to. Clipping at this stage is what keeps the whole thing tractable.
present_time	Year CE for the present-day fitting slice. Defaults to 1950, which is 0 BP and the present-day slice most reconstructions publish.
dataset_past	pastclim dataset for the palaeoclimate slices.
dataset_present	pastclim dataset for the present-day slice. Defaults to dataset_past, so one product is used throughout unless you deliberately ask for two. See details.
agg_past, agg_present	Aggregation factors; agg_present defaults to agg_past for the same reason.
present_dir	Subdirectory name for the present-day slice.
slice_fmt	sprintf() template for palaeoclimate subdirectory names.
check_grids	Verify that every written slice shares one grid.
quiet	Suppress progress messages.

Details

By default the present-day slice is drawn from the same product as the palaeoclimate slices. Mixing them – fitting on WorldClim while projecting onto CHELSA, as is easy to do by accident – folds the step between the two products into every `delta_from_present`, and that step is species-specific in sign, so it does not cancel. Overriding `dataset_present` is supported (an observational present has its own arguments in its favour) but it is now a deliberate act, and `check_climate_products()` will say so afterwards.

Where the two products differ in native resolution, set `agg_present` and `agg_past` so both land on the same cell size; `check_grids = TRUE` verifies that they did.

Existing files are skipped, so an interrupted run can simply be restarted.

Value

A `climate_dir()` source pointing at path, invisibly.

See Also

`climate_dir()`

Examples

```
## Not run:
prepare_climate(
  path = "climate/eurasia",
  times = seq(850, 1850, by = 100),
  extent = c(-15, 180, 10, 82)
)

## End(Not run)
```

presence_thresholds	<i>Alternative presence thresholds for each fitted model</i>
---------------------	--

Description

richcast binarises with the p10 threshold, which tolerates 10% omission of the training presences. Two alternatives, for reading presence as a band rather than a line, computed from each model's present-day suitability surface without refitting:

Usage

```
presence_thresholds(x, db)
```

Arguments

x	A <code>richcast_series</code> , a single <code>richcast_sdm</code> , or a list of them.
db	The <code>richcast_db</code> the models were fitted from.

Details

- `min_presence`: the lowest suitability the model assigns anywhere inside the species' own range polygon. The most permissive rule; suitability between it and `p10` reads as possible but marginal presence. It is set by the single worst cell in the range, so it is sensitive to ragged range edges.
- `tss`: the cutoff that maximises the true skill statistic (sensitivity + specificity - 1) when cells inside the range polygon are treated as presences and the rest of the study extent as absences. It depends on how wide the study extent is, and can fall above or below `p10`.

Value

A tibble with `species`, `threshold` (the fitted `p10` threshold), `min_presence` and `tss`.

See Also

[suitability_grid\(\)](#)

project_sdm

Project a fitted model onto a palaeoclimate slice

Description

Project a fitted model onto a palaeoclimate slice

Usage

```
project_sdm(sdm, climate, time, quiet = FALSE)
```

Arguments

<code>sdm</code>	A <code>richcast_sdm</code> from fit_sdm() .
<code>climate</code>	A climate source.
<code>time</code>	Year CE to project onto, or "present".
<code>quiet</code>	Suppress progress messages.

Value

A `richcast_projection`: the ensemble suitability surface over the species' study extent, the threshold, and the count of cells above it.

See Also

[fit_sdm\(\)](#)

region	<i>Define a modelling region</i>
--------	----------------------------------

Description

Either one of the preset regions, by name, or a user-defined box.

Usage

```
region(x, label = NULL)
```

Arguments

x	A preset name – one of "europe", "asia", "middle_east", "africa", "north_america", "south_america" – or a numeric box c(xmin, xmax, ymin, ymax) in decimal degrees. Case, spaces and hyphens in names are ignored, so "North America" works.
label	Optional name, used in output tables. Defaults to the preset name, or "custom" for a box.

Value

A richcast_region.

See Also

[region_presets](#)

Examples

```
region("europe")
region("Middle East")
region(c(68, 87, 39, 46), label = "Tian Shan")
```

region_presets	<i>Preset modelling regions</i>
----------------	---------------------------------

Description

Bounding boxes, as c(xmin, xmax, ymin, ymax) in decimal degrees, for the regions [region\(\)](#) accepts by name. They are deliberately generous and overlap where continents meet (the Middle East sits inside both Asia's and Africa's neighbourhoods); pass your own box to [region\(\)](#) for anything tighter.

Usage

```
region_presets
```

Format

A named list of numeric vectors of length 4.

Examples

```
names(region_presets)
region_presets$europe
```

richness_at

Predict richness, and the species expected, at coordinates

Description

Evaluates every fitted model at the given points, so the answer is exact at the climate grid's resolution and needs no stored surfaces. A species counts towards richness where its ensemble suitability is above its own threshold; expected_richness sums the suitabilities instead. The species table lists every modelled species at each point, most suitable first, which is the expected species list read straight off the suitability surface.

Usage

```
richness_at(x, lon, lat, time = "present", climate)
```

Arguments

x	A richcast_series, a single richcast_sdm, or a list of them.
lon, lat	Coordinates in decimal degrees, recycled to a common length.
time	Year(s) CE, and/or "present".
climate	The climate source the models were fitted with.

Details

A point outside a species' study extent is outside anything its model can speak to, so that species has NA suitability there and is not counted.

Value

A richcast_point with two tibbles: richness (one row per point per time) and species (one row per point, time and species).

See Also

[run_hindcast_series\(\)](#)

Examples

```
## Not run:
pt <- richness_at(res, lon = 35.5, lat = 33, time = c("present", 850),
                  climate = clim)

pt$richness
subset(pt$species, present)

## End(Not run)
```

richness_grid

*Richness surfaces as a tidy data frame***Description**

Long-format cell values for every slice, ready for a faceted map.

Usage

```
richness_grid(series, times = NULL, drop_na = TRUE)
```

Arguments

series	A richcast_series.
times	Which slices to include. Defaults to every hindcast slice; include "present" to add the present day.
drop_na	Drop cells no model covered.

Value

A tibble with x, y, time, richness and expected.

See Also

[richness_surface\(\)](#)

Examples

```
## Not run:
grid <- richness_grid(res)
ggplot2::ggplot(grid, ggplot2::aes(x, y, fill = richness)) +
  ggplot2::geom_raster() +
  ggplot2::facet_wrap(~time)

## End(Not run)
```

richness_in	<i>Predict richness, and the species expected, over a small focus area</i>
-------------	--

Description

The area counterpart of `richness_at()`. A single climate cell is often too small a unit to read an assemblage from, so this pools a handful of cells – say the 2 x 2 block covering one landscape – and asks which species the models place anywhere in it.

Usage

```
richness_in(x, area, time = "present", climate)
```

Arguments

x	A <code>richcast_series</code> , a single <code>richcast_sdm</code> , or a list of them. For a series, merged taxa are reported once, as in <code>run_hindcast_series()</code> .
area	The focus area, a <code>region()</code> box.
time	Year(s) CE, and/or "present".
climate	The climate source the models were fitted with.

Details

Every grid cell whose centre lies inside `area` is evaluated. A species is **present** in the area when its ensemble suitability clears its threshold in at least one of those cells, and its suitability is the highest value among them, so the two agree: present exactly when `suitability > threshold`. `richness` counts the species present; `expected_richness` sums their suitabilities. Cells without climate (sea, or ice) are skipped, so an area on a coast can hold fewer cells in some slices than others – `cells` records how many were evaluated.

Value

A `richcast_area` with two tibbles: `richness` (one row per time) and `species` (one row per time and species, with `suitability`, `threshold`, `present` and `cells_present`).

See Also

`richness_at()`, `run_hindcast_series()`

Examples

```
## Not run:
focus <- region(c(34.5, 35.5, 31.5, 32.5), label = "focus area")
fa <- richness_in(res, focus, time = c("present", -4050), climate = clim)
fa$richness

## End(Not run)
```

richness_surface	<i>Extract a richness surface from a series</i>
------------------	---

Description

Extract a richness surface from a series

Usage

```
richness_surface(series, time, layer = c("richness", "expected", "both"))
```

Arguments

series	A richcast_series from run_hindcast_series() .
time	Year CE, or "present".
layer	"richness" for the count of species above threshold, "expected" for the sum of suitabilities, or "both".

Value

A SpatRaster.

See Also

[richness_grid\(\)](#), [run_hindcast_series\(\)](#)

rodent_traits	<i>Rodent ecological and zoonotic trait table</i>
---------------	---

Description

Species-level attributes for 2000+ rodent species, from Supplementary Data 1 of Ecke et al. (2022). Bundled as a worked example of the trait table that [build_taxon_db\(\)](#) joins to a range source; there is nothing rodent-specific about the rest of the package.

Usage

```
rodent_traits
```

Format

A tibble with one row per species. Columns of most interest:

species Binomial in Genus_species form (normalised by `normalise_species()`).

S_index Synanthropy index. Note that every species present in this table scores above zero, so $S_index > 0$ is a presence-in-table filter rather than a threshold.

Synanthropic Binary synanthropy flag.

Hunted Binary hunting-pressure flag.

Reservoir Binary zoonotic-reservoir flag.

NA_m North America occurrence flag. Renamed from the source column NA, which collides with R's missing-value sentinel.

Remaining columns carry habitat use, zoonosis counts by pathogen group, population-dynamics measures, continental occurrence and MSW05 taxonomy. See the source publication for full definitions.

Source

Ecke, F., Han, B. A., Hornfeldt, B., Khalil, H., Magnusson, M., Singh, N. J., & Ostfeld, R. S. (2022). Population fluctuations and synanthropy explain transmission risk in rodent-borne zoonoses. *Nature Communications*, 13, 7532. doi:10.1038/s41467022352737. Redistributed under CC BY 4.0.

Examples

```
head(rodent_traits[, c("species", "S_index", "Synanthropic", "Hunted")])
attr(rodent_traits, "source")
```

run_hindcast_series	<i>Hindcast a whole assemblage across a series of time slices</i>
---------------------	---

Description

Selects the species near region with `species_near()`, fits one ensemble model per species with `fit_sdm()`, projects each onto every requested time slice, and stacks the results into richness surfaces for the region.

Usage

```
run_hindcast_series(
  db,
  climate,
  times,
  region,
  species = NULL,
  merge = NULL,
  keep_surfaces = TRUE,
```

```

    on_error = c("warn", "stop"),
    quiet = FALSE,
    ...
)

```

Arguments

db	A richcast_db from <code>build_taxon_db()</code> .
climate	A climate source from <code>climate_dir()</code> or <code>pastclim_climate()</code> .
times	Numeric vector of years CE.
region	The modelling region, from <code>region()</code> .
species	The taxa to model. NULL (the default) uses the Pleistocene zooarchaeological list for the preset region (<code>zooarch_taxa()</code>); otherwise a character vector of your own. Either way each species is narrowed by <code>species_near()</code> : a species more than 10 degrees from region is never trained.
merge	Taxa that count as one identification, as a named list: <code>list(Dama_sp = c("Dama_dama", "Dama_mesopotamica"))</code> . A merged taxon named in species has each member modelled separately, on its own range, and the outputs combined: the taxon is present in a cell when any member is, and its suitability there is that of the member furthest above its own threshold. Modelling the union of the ranges instead lets the larger range swamp the smaller one. NULL (the default) uses the merges the region's zooarchaeological list defines.
keep_surfaces	Retain the richness surfaces, so maps can be drawn afterwards. Stored wrapped, so the result survives <code>saveRDS()</code> .
on_error	"warn" skips a failing species and carries on; "stop" aborts the run.
quiet	Suppress per-species progress.
...	Further arguments passed to <code>fit_sdm()</code> , e.g. predictors or n_presence.

Details

Each species is fitted **once**, against present-day climate over its own study extent, then projected repeatedly. Training is independent of the region: the region decides only which species are trained and where richness is summarised.

Each slice gets two surfaces on the climate grid cropped to the region: richness, the number of species above their own threshold in each cell, and expected, the sum of the species' ensemble suitabilities – a threshold-free estimate of how many species a cell supports.

Value

A richcast_series:

- richness: one row per slice (present first) with mean, median and maximum richness over the region, and mean_expected.
- species: one row per taxon per slice. cells counts suitable cells over the study extent (summed over members for a merged taxon), region_cells over the region, each with its change from the present.

- `models`: one row per fitted species, with the taxon it belongs to, the fit diagnostics and the AUC and Boyce index of each ensemble member and of the ensemble.
- `fits`: the fitted models, for `project_sdm()` and `richness_at()`.
- `taxa`: which fitted species make up each taxon.
- `surfaces`: the richness surfaces; see `richness_surface()`.

See Also

`fit_sdm()`, `richness_at()`, `richness_surface()`

`sf_polygons`

Range source: arbitrary sf polygons

Description

The general-purpose backend. Use it for any polygon range map you already hold in R – a national atlas, expert-drawn ranges, alpha hulls, or the output of someone else’s SDM.

Usage

```
sf_polygons(x, species_col = "species")
```

Arguments

`x` An sf object with polygon or multipolygon geometry.

`species_col` Name of the column holding species names.

Value

A `richcast_range_source`.

Examples

```
ranges <- sf::st_sf(
  species = c("Genus_alpha", "Genus_beta"),
  geometry = sf::st_sfc(
    sf::st_polygon(list(cbind(c(0, 2, 2, 0, 0), c(0, 0, 2, 2, 0)))),
    sf::st_polygon(list(cbind(c(1, 3, 3, 1, 1), c(1, 1, 3, 3, 1)))),
    crs = 4326
  )
)
sf_polygons(ranges)
```

species_near	<i>Species whose ranges lie near a region</i>
--------------	---

Description

The species richcast will train for a region: those whose present-day range polygon comes within 10 degrees of the region's box. The distance is fixed, and `run_hindcast_series()` always applies it, so a species far from the region can never enter its richness – a model extrapolated to a continent the species has never occupied will happily place a llama in Britain.

Usage

```
species_near(db, region, quiet = FALSE)
```

Arguments

db	A richcast_db.
region	A <code>region()</code> .
quiet	Suppress the report.

Details

Distance is measured on the box expanded by 10 degrees in longitude and latitude, which is quick and conservative near the poles.

Value

A character vector of species names.

Examples

```
db <- build_taxon_db(
  sf::st_sf(
    species = c("Genus_near", "Genus_far"),
    geometry = sf::st_sfc(
      sf::st_polygon(list(cbind(c(0, 1, 1, 0, 0), c(0, 0, 1, 1, 0)))),
      sf::st_polygon(list(cbind(c(60, 61, 61, 60, 60), c(0, 0, 1, 1, 0)))),
      crs = 4326
    )
  ),
  quiet = TRUE
)
species_near(db, region(c(5, 10, 0, 5)))
```

suitability

Extract the suitability surface from a fitted model or projection

Description

Rasters are stored wrapped so that objects survive `saveRDS()`; this unwraps one back into a usable `SpatRaster`.

Usage

```
suitability(x, binary = FALSE)
```

Arguments

<code>x</code>	A <code>richcast_sdm</code> or <code>richcast_projection</code> .
<code>binary</code>	Return presence (1) / absence (0) at the model's threshold instead of continuous suitability.

Value

A `SpatRaster`.

suitability_grid

Suitability of every fitted species in every cell of an area

Description

The cell-level table behind `richness_in()`: each grid cell whose centre lies inside area, evaluated by every fitted model whose study extent covers it. Cells without climate are skipped. Merged taxa are not collapsed here – each fitted species keeps its own rows – so thresholds other than the fitted ones can be applied afterwards.

Usage

```
suitability_grid(x, area, time, climate)
```

Arguments

<code>x</code>	A <code>richcast_series</code> , a single <code>richcast_sdm</code> , or a list of them. For a series, merged taxa are reported once, as in <code>run_hindcast_series()</code> .
<code>area</code>	The focus area, a <code>region()</code> box.
<code>time</code>	A single year CE, or "present".
<code>climate</code>	The climate source the models were fitted with.

Value

A tibble with x, y, species, suitability and threshold, one row per cell and fitted species. The number of cells with climate is attached as the cells attribute.

See Also

[richness_in\(\)](#), [presence_thresholds\(\)](#)

zooarch_taxa	<i>Pleistocene zooarchaeological species lists</i>
--------------	--

Description

The species richcast models by default for a preset [region\(\)](#): taxa reported from Pleistocene zooarchaeological or palaeontological sites in that region, restricted to species that have present-day IUCN range maps. Extinct taxa (aurochs, *Equus hydruntinus*) and taxa outside the bundled families have no range polygon to train on and are left out.

Usage

```
zooarch_taxa(region = NULL)
```

Arguments

region A [region\(\)](#) or a preset name. NULL returns every list.

Value

A tibble with region, species, evidence, source and members (for a merged taxon, the IUCN species it combines).

Middle East

Nine taxa of bovids, cervids, equids and suids from the Levant, the Zagros, Syria and Arabia, compiled from:

- Stewart, M., Louys, J., Price, G. J., Drake, N. A., Groucutt, H. S., & Petraglia, M. D. (2019). Middle and Late Pleistocene mammal fossils of Arabia and surrounding regions: implications for biogeography and hominin dispersals. *Quaternary International*, 515, 12-29.
- Stiner, M. C. (2005). *The Faunas of Hayonim Cave, Israel*. Peabody Museum, Harvard.
- Yeshurun, R., Bar-Oz, G., & Weinstein-Evron, M. (2007). Modern hunting behavior in the early Middle Paleolithic: faunal remains from Misliya Cave, Mount Carmel, Israel. *Journal of Human Evolution*, 53, 656-677.
- Stiner, M. C., Barkai, R., & Gopher, A. (2009). Cooperative hunting and meat sharing 400-200 kya at Qesem Cave, Israel. *PNAS*, 106, 13207-13212.
- Tsahar, E., Izhaki, I., Lev-Yadun, S., & Bar-Oz, G. (2009). Distribution and extinction of ungulates during the Holocene of the southern Levant. *PLoS ONE*, 4, e5316.

- Mata-Gonzalez, M., Starkovich, B. M., Zeidi, M., & Conard, N. J. (2023). Evidence of diverse animal exploitation during the Middle Paleolithic at Ghar-e Boof (southern Zagros). *Scientific Reports*, 13, 19006.

Older names are mapped to current IUCN taxonomy: *Capra ibex* from the Negev to *Capra nubiana*, and *Ovis orientalis* to *Ovis gmelini*.

Species that are not reliably told apart in the record are one taxon, with the members column naming the species it combines: *Dama_sp* (*Dama dama*, *D. mesopotamica*), *Gazella_sp* (*Gazella gazella*, *G. subgutturosa*, *G. dorcas*), *Capra_sp* (*Capra aegagrus*, *C. nubiana*) and *Large_antelope* (*Alcelaphus buselaphus*, *Oryx leucoryx*). [run_hindcast_series\(\)](#) models each member on its own range and then combines them. *Equus ferus*, *E. africanus* and *Gazella arabica* are reported in the sources but left out on review. Wild boar needs its own IUCN range polygon alongside the ungulate downloads.

Lists for the other presets are not yet compiled; pass your own species to [run_hindcast_series\(\)](#) for those regions.

Examples

```
zooarch_taxa("middle_east")
```